PERCENTAGE ALL TYPES

Date

Concepts 1:-

(1) 40 is 20% of a number
What is 80% of that no.?

$$40 = \frac{20}{100} \times x$$

$$x = 200$$

$$200 \times \frac{80}{100} = 160$$

Unitary method

$20\% \rightarrow 40$

$1\% \rightarrow \frac{40}{20} \times 80 = 160$

Module 1 : Basic Percentage

Concept : Difference between 54% of a number & 63% of the same number is 72. What is 80% of that number?

$9\% — 72$

$1\% — \frac{72}{9} \times 80$

640

1. Sameer spends 20% of his monthly salary on house-rent, 25% on food, 10% on transportation, 15% on education of his children and 18% on household expenses. He saves the remaining amount of 4800. What is his monthly salary? 1) 40000 2) 38000 3) 32000 4) 36000

5) None of these.

$12 — 4800$

$1 — \frac{4800}{12} \times 100$

Ans = 40000

Date

2. Ishan spent 35,645 on buying a bike, 24,355 on buying a television, and the remaining 20% of the total amount he had as cash with him. What was the total amount?

1) 60,000 2) 72,000 3) 75,000 4) 80,000
5) None of these.

20% of the total amount he had as cash with him

∴ Total expenditure = 80%

$80\% = 60,000$ ~~30,000~~ 15,000

$1\% = \frac{60,000}{80} \times 100$

$1\% = 75,000$

3. Sujatha invests 7% i.e. 2170, of her monthly salary in mutual funds. Later she invests 18% of her monthly salary in recurring deposits. Also she invests 6% of her salary on NSC's. What is the total annual income invested by Sujata?

1) 125320 2) 113520 3) 1351.20 4) 115320
5) None of these

$7\% = 2170$

$1\% = \frac{2170}{7} \times 31 \times 12$

$9160 \times 12 = 113320$

Date

4. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount of 12750. What is Sonali's monthly income?

1) 42500 2) 38800 3) 40,000 4) 35,000 5) None of these.

70

~~1650~~
~~12750~~ ×100
~~70~~

~~16500~~ 425
~~30~~ ~~12750~~ × 100
~~30~~

$\boxed{1)\ 42500}$

5. Gaurav spends 40% of the amount he received from father on hostel expenses, 20% on books and stationery and 50% of the remaining on transport. He saves 450 which is half the remaining amount after spending on hostel.

1) 3000 2) 6000 3) 4500 4) 5000 5) None of these

20 = 450

1 = $\frac{450}{20}$ × 100

2250 is half

2250
+2250
4500

or

20% = 900

= $\frac{900}{20}$ × 100

= 4500

450
450
5
2250
2250
4500

Date

6. Prerna decided to donate 30% of her monthly income to orphanage. But on the day of donation, she changed her mind and donated ₹4200, which is 70% of what she has decided earlier. What should be the actual donation as per her earlier decision?

1) 8000 2) 5000 3) 6,000 4) cannot be determined 5) None of these.

$70 = 4200$

$1 = \frac{4200}{70} \times 100$ 60

$= 6000$

~~PRACTICE: AC 18 What percentage~~

7. Praveen decided to donate 15% of his salary to a charity. On the day of donation he changed his mind and donated ₹1896, which was 80% of what he had decided earlier. How much is Praveen's salary?

1) 18500 2) 10,250 3) 15,800 4) cannot be determined 5) None of these.

$80 = 1896$ 2367

$1 = \frac{1896}{80} \times 100$

Let Praveen's salary $= x$

Date.....

Step 1 — Original donation

He decided to donate 15% of Salary
Original donation $= 0.5x$

Step 2 — Changed donation

He donated ₹1896, which is 80% of the original donation.

$$0.8 \times 0.15x = 1896$$
$$0.12x = 1896$$

Step 3 — Find Salary

$$x = \frac{1896}{0.12}$$

$$x = 15800$$

Final answer :- ₹15,800